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Structures of Cu_n^+ (n = 3−10) Clusters Obtained by Infrared Action Spectroscopy

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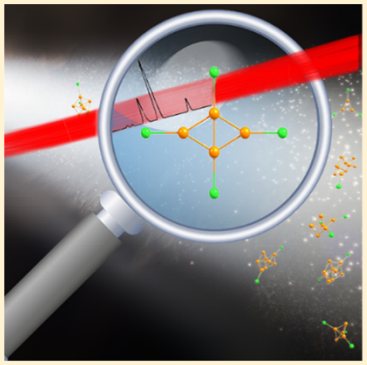
Supporting Information

ABSTRACT: Coinage metal clusters are of great importance for a wide range of scientific fields, ranging from microscopy to catalysis. Despite their clear fundamental and technological importance, the experimental structural determination of copper clusters has attracted little attention. We fill this gap by elucidating the structure of cationic copper clusters through infrared (IR) photodissociation spectroscopy of Cu_n^+−Ar_m complexes. Structures of Cu_n^+ (n = 3−10) are unambiguously assigned based on the comparison of experimental IR spectra in the 70−280 cm^{-1} spectral range with spectra calculated using density functional theory. Whereas Cu_3^+ and Cu_4^+ are planar, starting from n = 5, Cu_n^+ clusters adopt 3D structures. Each successive cluster size is composed of its predecessor with a single atom adsorbed onto the face, giving evidence of a stepwise growth.



Transition-metal clusters are of great interest and considerable importance to the fields of heterogeneous catalysis, solid-state physics, surface chemistry, and organometallic chemistry due to their frequently strongly size-dependent properties of magnitudes that are often not present in small molecular or bulk systems. Part of the fascination of clusters stems from their often enhanced reactivity with respect to the bulk, typically attributed to the larger number of undercoordinated atoms, involving free d electrons to actively participate in the bonding. The coinage metals' (Cu, Ag, Au) special place in this stems from their closed d shell, forcing s electrons to become involved in bonding and often leading to much more gentle activation of feedstock molecules.[1,2]

remains elusive, even for the very first step. Using clusters as well-defined model systems, proposed reaction pathways can be tested with high confidence, allowing much needed insight. Beside this role as a model system, clusters may also act as a catalyst under technical conditions themselves. Recently, it was shown that deposited Cu_4^+ can lower the activation barrier for the methanol formation from CO_2,[7] thereby illustrating the potential of nanotailored catalysts.

For both roles, however, knowledge of the clusters themselves and, in particular, of their structures, is imperative. In the model system scenario, it is of great importance to know the cluster morphology prior to exposing it to reactants, whereas for studies on deposited clusters, size-selected prior to

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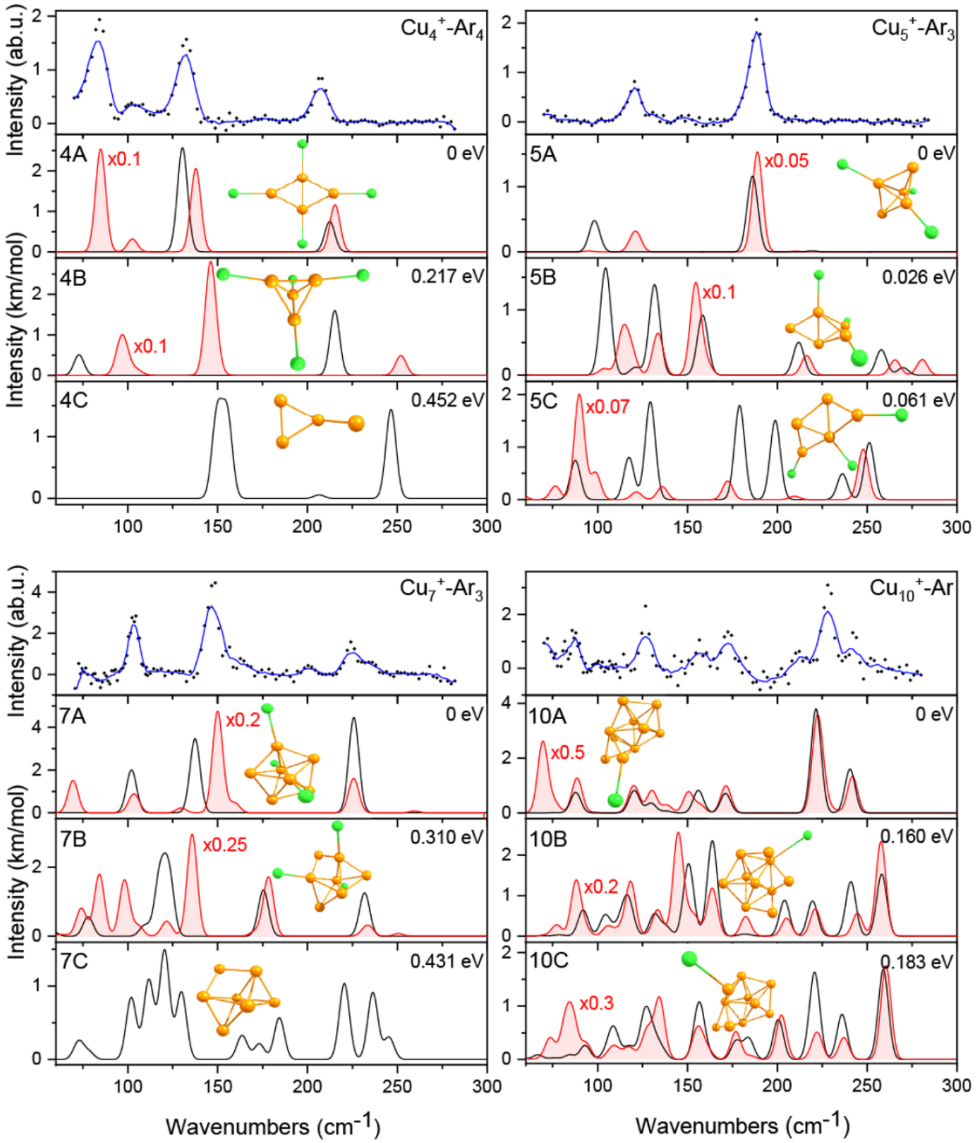

Figure 1. Comparison of IRMPD spectra of Cu_4^+−Ar_4, Cu_5^+−Ar_3, Cu_7^+−Ar_3, and Cu_{10}^+−Ar (black dots; the blue line represents a five-point adjacent average) with calculated vibrational spectra of three low-energy isomers (black lines) and their complexes with Ar (red shading). The corresponding structures and relative energies of the bare clusters are reported above each spectrum.

Infrared multiple photon dissociation (IRMPD) spectroscopy has a solid track record in determining the molecular structure of metal clusters and can do so mass-selectively using weakly bound messenger atoms or molecules.[18,19] We recorded IRMPD spectra of Cu_n^+−Ar_m (n = 3−10, m = 1−4) in a molecular beam environment using IR light produced by the compromise between low Ar coverage and the mass range over which IR spectra could be recorded. To reduce contamination due to ingrowth from Cu_n^+−Ar_{m+1} complexes, for each cluster size n, we display the spectrum for the complex Cu_n^+−Ar_m with the highest intensity in the mass spectrum. As a result, spectra for the lower cluster masses have been recorded at relatively

that each of the structures proposed does not have sufficient bands to explain the experimental spectrum. Sure enough, the 132 and 207 cm^{-1} bands match the calculated spectrum for the lowest energy D_{2h} rhombic structure 4A very well, but this leaves the lower frequency bands unaccounted for. The inclusion of the Ar messenger atoms in the DFT calculations quite drastically changes the predicted spectrum. Now all four observed bands are predicted, agreeing in both frequency and relative intensity. The two lowest-frequency bands (observed at 83 and 103 cm^{-1}) that were not predicted by the calculations for the bare Cu_4^+ originate from motions of the Ar relative to cluster. Also, slight shifts for the higher frequency bands are calculated, from 130 and 212 cm^{-1} for the bare clusters to 138 and 215 cm^{-1} for the complex. These bands are associated with elongations of the cluster along the Cu−Cu axes, which are sterically hindered by the presence of the Ar. However, they do not disturb the original structure dramatically, as illustrated by the relatively low calculated binding energies of 0.20 and 0.27 eV for binding to the copper atoms positioned along the long and short axes, respectively. It is of further interest that the presence of Ar atoms increases the IR intensity of the bands to over ten times, suggesting that the Ar acts as an "antenna".

The IRMPD spectrum recorded for Cu_5^+−Ar_3 is simpler than that for Cu_4^+−Ar_4, exhibiting just two well-resolved bands and suggesting a higher symmetry structure. In the literature, the structure of Cu_5^+ has been the subject of considerable debate, with both planar and 3D structures proposed to be most stable.[2,17,22,23] Here it is compared with the calculated spectra of three isomers of Cu_5^+ within 0.1 eV relative to each other. The comparison for the bare structures suggests the assignment to a D_{3h} trigonal bipyramid isomer 5A, with a reasonable agreement for the bands observed at 188 cm^{-1} (177 cm^{-1} calculated) and at 121 cm^{-1} (93 cm^{-1} calculated). Calculated modes correspond to in-plane and out-of-plane distortions of the triangular base plane. Ar binding to atoms of the triangular base is energetically the most favorable, where the steric hindrance caused by the Ar shifts the calculated bands to 189 and 121 cm^{-1}, respectively, resulting in an almost perfect match with the experimental bands. Ar complexation of the planar structure 5C is not energetically favorable (0.06 eV higher than the lowest energy complex of 5A), nor does it provide a better agreement with the observed spectrum. IMS measurements by Weis suggest a planar structure 5C for Cu_5^+.[17] Because the current calculations suggest that Ar complexation energetically favors structure 5A, the current work cannot decisively conclude the discussion on the structure of Cu_5^+.

The IRMPD spectrum of Cu_7^+−Ar_3 shows four bands at

essentially the same as that for the bare structure, except for some shifting bands that do not match the experimental spectrum. For 7A, the presence of two Ar atoms on the five-fold symmetry axis does not lead to symmetry breaking, but the third one in the pentagonal plane does, albeit only slightly. As a consequence, extra bands appear in the spectrum, for instance, the high-frequency shoulder of the main band at 150 cm^{-1}, which explains the asymmetric line shape of the observed band. A hint of a band near 70 cm^{-1} is observed, which would agree with the cluster−Ar vibration predicted. However, because the FEL power is lowest here, a power correction may amplify background noise, and because we do not see the potential band return to the baseline, we cannot draw firm conclusions.

The final system shown is Cu_{10}^+. With the increase in cluster size, the number of absorbed Ar atoms decreases due to the dilution of local charge density. For this reason, structure determination of the Cu_{10}^+ cluster could be done by employing the photodissociation spectrum of Cu_{10}^+−Ar. The spectrum reflects the growing complexity of the system, showing a rich structure with many overlapping bands. Nevertheless, even on a first comparison with calculated spectra, we are able to assign it to structure 10A, a pentagonal bipyramid, now capped with three Cu atoms on neighboring faces, making it also interpretable as two interlocked pentagonal bipyramids that share four atoms. With the exception of a weaker band at 213 cm^{-1}, all spectral features are matched. Whereas the 213 cm^{-1} band could be due to the presence of another isomer, we conclude that the spectrum is by and large dominated by structure 10A. Because of the increase in the system size, the relative influence of the Ar messenger atom on the spectrum is further reduced in comparison with the Cu_7^+ case: DFT calculations for the Ar-tagged system predict an almost identical spectrum as that for the bare cluster, with (apart from a cluster−Ar vibration at 70 cm^{-1}) minimal modifications in band frequencies and only a two-fold intensity increase.

In this way, we recorded IRMPD spectra for Cu_n^+ (n = 3−10) and unambiguously assigned each spectrum to a specific structure. Further experimental and calculated spectra, including spectra for Ar-complexed clusters, can be found in the SI. Beside the clusters discussed above, we find a triangular structure for Cu_3^+, a capped triangular bipyramid structure for Cu_6^+, and a pentagonal bipyramid structure capped with one and two Cu atoms for Cu_8^+ and Cu_9^+, respectively. It is now of interest to evaluate the stepwise evolution of the cluster structures found, as shown in Figure 2. Clearly, the two smallest clusters Cu_3^+ and Cu_4^+ are planar. However, from n = 5, all structures found are 3D. What is particularly interesting is that every structure found is formed by the addition of a Cu

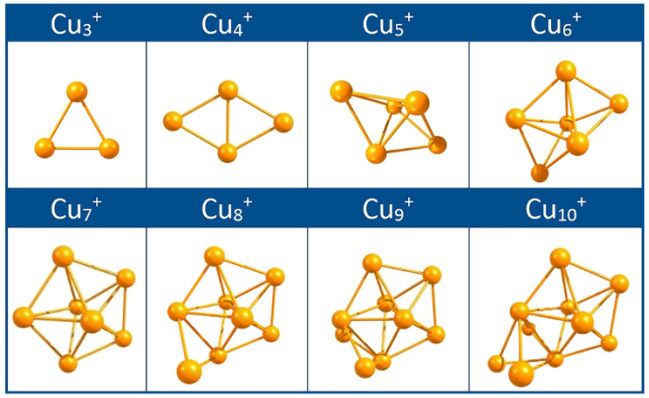

Figure 2. Structures of the small cationic copper clusters (n = 3−10) inferred in this work.

charge for the cluster geometry: The structure is a fine balance between interatomic stabilization and surface energies, and as a consequence, the loss of an electron leads to the weakening of the interatomic bonds, and the cluster needs to rearrange to reduce the surface energy.[22]

We can further compare the structures found for Cu_n^+ with those for other coinage metals. Interestingly, for Ag_n^+, the same structures for the small cluster sizes (n = 3−10) are found, except for n = 5.[35] It is further noteworthy that the structure found for Cu_4^+ is the same as that used for computational studies on deposited Cu_4^+ clusters. Such calculations are thus consistent with the form the clusters had prior to deposition on the Al_2O_3 support.[7]

In conclusion, we established the structure of small cationic copper clusters based on a combination of IR photofragmentation spectroscopy of Cu_n^+−Ar_m complexes and DFT calculations, which convincingly account for the influence of the Ar.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpclett.9b00539.

Experimental and computational details, IRMPD spectra of clusters with n = 3−10 and their comparison to different bare isomers, and the isomers of Ar-tagged complexes of the matching isomer (PDF)

■ AUTHOR INFORMATION

■ ACKNOWLEDGMENTS

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